SADMAN SAKIB

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Technical Summary

- Research experience in Human-Computer Interaction, including semi-structured interviews and thematic analysis.
- Research experience in developing a computer vision based smart wearable system using Raspberry Pi and sensors for obstacle avoidance, fall detection, and real-time OCR.
- Research experience in computer vision for object detection, currency classification, and face recognition.
- Research experience in human-centered AI and accessibility for the visually impaired and people who stutter.
- Developed a mobile app for glucose and hemoglobin estimation from fingertip videos using PPG and deep learning.
- Developed a query-aware image paneling approach to enhance long-video understanding with vision language models.
- Developed a graph neural network (GNN) and BERT fusion pipeline for multimodal music understanding.
- Hands-on experience with deep learning frameworks and libraries, including PyTorch, Keras, and Hugging Face.
- Hands-on experience in deploying deep learning models on resource constrained edge devices such as Raspberry Pi.
- Hands-on experience working on a leading fintech platform built on microservice-based architecture.

Education

M.Sc. in CSE at BRAC University **Jan. 2026 – June 2027 (Exp.)**
12 credits completed *Dhaka, Bangladesh*

- CGPA: 4.00/4.00, up to 2nd semester.

B.Sc. in CSE at Khulna University of Engineering & Technology (KUET) **Feb. 2019 – March 2024**
Thesis: An Assistive Device for the Visually Impaired for Enhanced Navigation and Reading *Khulna, Bangladesh*

- CGPA: 3.73/4.00

Skills

Programming Languages: Python, C, C++, Java.
Machine Learning: PyTorch, Hugging Face, NumPy, Pandas.
Computer Vision: OpenCV, YOLO, Tesseract OCR, CLIP.
Developer Tools: Anaconda, Jupyter, Raspberry Pi, Git, VS Code, IntelliJ IDEA, Postman.
Mobile Apps Development: Kotlin, Android Studio.
Backend: Spring Boot, Kafka, RabbitMQ.
Database: MySQL, Oracle, Firebase, Redis.
Operating System: Linux, Windows.
Industry Expertise: Microservices Architecture, Fintech Platforms, Digital Payment Systems.
Research Aptitude: Human-Computer Interaction, Accessibility and Assistive Technology, Computer Vision.
Adaptive Skills: Adaptability, Analytical Thinking, Clear Technical Communication, Complex Problem Solving, Cross-Functional Collaboration, Independent Learning.

Experience

Software Engineer **Aug. 2024 – Aug. 2026**
Kona Software Lab Ltd. *Dhaka, Bangladesh*

- Developed backend services for fintech and mobile financial service (MFS) platforms using a microservices architecture.
- Implemented core wallet transaction flows and bulk disbursement services handling high-volume financial transactions.
- Integrated payment services and systems, including mastercard payment gateway services for card transactions, a central payment network switch for interbank transfers, and QR code for interoperable QR-based merchant payments.
- Designed system solutions with client teams and cross-functional stakeholders, from requirements analysis to delivery.

Projects

Query-Guided Grid Composition for Long Video Understanding | *Python, PyTorch, VLMs* **Feb. 2026 – Present**

- Developed a training-free method that helps vision–language models answer questions about long videos.
- Selected query-relevant frames using CLIP similarity, spread across the video timeline so no single scene dominates.
- Sized each panel by its relevance to the query, giving more relevant frames a larger area and higher resolution.
- Evaluated with Qwen2.5-VL-7B on Video-MME and MF2, improving MF2 pair accuracy from 16.9% to 23.5%..

AI-Assisted Video Creation for People Who Stutter | *HCI, Qualitative Research* **May 2026 – Sep. 2026**

- Explored how people who stutter in Bangladesh create videos to share their voices and represent their community, through an onboarding workshop and semi-structured interviews.
- Used reflexive thematic analysis to reveal how audience and context shape their choices to keep or edit out stuttering, and what they need from AI editing tools.

GNN-BERT for Music Context Understanding | *PyTorch, PyTorch Geometric, BERT* **Jan. 2026 – May 2026**

- Fine-tuned BERT for multi-label music tagging from natural-language captions.
- Modeled music as graphs of audio segments and classified genre with a GraphSAGE graph neural network.
- Fused graph-based audio and BERT text embeddings using cross-attention for genre and mood prediction,improving AUC-PR by 22% over the best single modality.
- Learned a shared audio–text representation with contrastive learning for cross-modal retrieval and zero-shot tagging.

Assistive Smart Spectacle for the Visually Impaired | *OpenCV, Pytesseract, Raspberry Pi* **May 2023 – May 2024**

- Built a Raspberry Pi based spectacle that alerts users to obstacles using laser sensors and notifies a caregiver when a fall is detected.
- Built an OCR pipeline using image processing techniques, with text-to-speech output.
- Achieved 6.97% character error rate and 9.46% word error rate in OCR.

Automated Object Identification for the Visually Impaired | *PyTorch, YOLOv5, OpenCV* **July 2023 – June 2024**

- Trained a YOLOv5 detector on a custom dataset to recognize common indoor objects such as chairs, doors, stairs, and people in real time, achieving 98.2% mean Average Precision (mAP).
- Built a cascaded pipeline that crops detected banknotes for currency classification, achieving 99.7% top-1 accuracy.
- Implemented face recognition by matching ResNet-50 embeddings using cosine similarity.

Non-Invasive Glucose and Hemoglobin Measurement App | *Kotlin, Python, Firebase* **Dec. 2022 – May 2023**

- Developed an Android application that estimates glucose and hemoglobin levels from a fingertip video recorded with the phone camera and flash.
- Extended CameraX API to control ISO, exposure, and brightness, keeping video quality consistent across various lighting conditions.
- Extracted PPG signals from the video and integrated a deep learning model to predict blood component levels, easing the painful experience of blood sampling.

Vikings Chess | *Python, Pygame* **Feb. 2023 – May 2023**

- Developed a turn based board game where the opponent is an Artificial Intelligence based system.
- Built an AI opponent using minimax search algorithm with alpha-beta pruning, and designed a heuristic evaluation function to choose its moves efficiently

Relevant Coursework

- Artificial Intelligence
- Neural Networks and Fuzzy Systems
- Deep Learning for Computer Vision
- Natural Language Processing
- Research Methodology (HCI focused)
- Fairness, Ethics and Explainability in AI
- Image Processing
- Embedded Systems
- Data Mining
- Applied Statistics
- Mobile Computing
- Computer and Network Security

Publications

- **S. Sakib**, M. S. Sadi, M. S. Arefin, M. R. Islam, and H. Neha, "An Assistive Device for the Visually Impaired for Enhanced Navigation and Reading," 13th International Conference on Electrical and Computer Engineering (ICECE), Dhaka, Bangladesh, 18-20 Dec., 2024, pp. 767-772, doi: 10.1109/ICECE64886.2024.11024586.
- H. Neha, **S. Sakib**, F. Sadaf, and S. M. T. U. Raju, "Mobile Application to Collect Data and Measure Blood Component Level in a Non-Invasive Way," 26th International Conference on Computer and Information Technology (ICCIT), Cox's Bazar, Bangladesh, 13-15 Dec., 2023, pp. 1-6, doi: 10.1109/ICCIT60459.2023.10441186.
- H. Neha, M. S. Sadi, M. R. Islam, and **S. Sakib**, "Automated Object Identification for the Visually Impaired," 27th International Conference on Computer and Information Technology (ICCIT), pp. 3584-3589, Cox's Bazar, Bangladesh, 20-22 Dec., 2024, doi: 10.1109/ICCIT64611.2024.11022362.

Honors & Activities

- Dean's Award for excellent academic performance at KUET for academic years 2019–20, 2020–21, and 2021–22.
- Solved over 1,100 algorithmic problems on online judges such as Codeforces, AtCoder, CodeChef, and LeetCode, reaching the Specialist rank on Codeforces (max rating 1442).
- Special Group Interested in Programming Contest (SGIPC), KUET: organized programming workshops to train students for competitive programming contests.